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Research

Analytical framework for UAV-enabled wireless communication with D2D networks

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Abstract
This study analyzes the performance of Device-to-Device (D2D) communication in drone-assisted cellular networks, focusing on coverage probability, system throughput, and energy efficiency. Simulation results reveal that increasing UAV altitude improves the probability of line-of-sight (LoS) communication, enhancing coverage probability up to 92% at optimal altitudes. However, higher altitudes also increase path loss, leading to a 15% reduction in total system throughput. Additionally, an optimal D2D user density of 50 users per km^2 is identified to balance interference and resource utilization effectively. Uplink and downlink scenarios show that interference significantly affects the success transmission probability, which declines by 25% when the SINR threshold increases from 0 to 10 dB. In public safety scenarios, underlay in-band communication enables D2D users to establish reliable indirect links via cellular users, ensuring critical message delivery with energy efficiency improvements of up to 18%. These findings provide actionable insights for optimizing UAV deployment and resource allocation in D2D networks.

Article Highlights

- UAV altitude boosts LoS, achieving 92% coverage, but increases path loss, reducing throughput by 15%
- SINR rise from 0 to 10 dB lowers success transmission probability by 25%, requiring better interference management.
- Underlay in-band D2D links enhance public safety, ensuring message delivery with 18% energy efficiency improvement

Keywords UAV · Device-to-device communication · Public safety · Energy efficiency

1 Introduction

The surge in mobile devices has increased data traffic, creating a need for enhanced wireless network capacity. High-speed data transmission in dense areas is challenging, but integrating Device-to-Device (D2D) communication with mobile networks and Unmanned Aerial Vehicles (UAVs) as flying base stations offers a solution [1]. UAVs enhance network capacity and coverage through line-of-sight (LoS) communication and mobility, allowing rapid deployment to improve Quality of Service (QoS) [2, 3]. This approach supports scalability and efficiency, addressing

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the demands of dense environments while improving network performance and user connectivity. UAVs, or drones, enhance wireless communication with (LoS) connectivity, improving data rates and network coverage. They are widely used in military, surveillance, telecommunications, and rescue operations [4, 5]. UAVs, classified by altitude into HAPs (above 17 km) and LAPs (a few km), serve diverse applications. HAPs excel in endurance, while LAPs offer rapid deployment for emergencies. Fixed-wing UAVs are faster and heavier; rotary-wing UAVs hover effectively. Proper deployment ensures cost-effective, reliable wireless communication solutions [6, 7]. UAVs support disaster management by acting as aerial base stations during infrastructure failures, ensuring reliable communication in emergencies like floods or earthquakes [8, 9]. D2D communication and UAVs improve wireless networks by enhancing coverage and reducing interference. UAVs act as mobile base stations, enabling efficient, direct connections in dense areas, while D2D reduces network traffic. Together, they optimize connectivity, expand capacity, and ensure low-latency, reliable services. D2D communication, defined in 3GPP Release 12, plays a vital role in 5G for social networks, commercial services, and public safety, enhancing local data transfer and wireless network coverage [10–12].

Recent research has focused on modeling air-to-ground (A2G) communication channels for unmanned aerial vehicles (UAVs). Zhang et al. proposed a novel neural network model designed to predict path loss in air-to-ground (A2G) communication links. The model leverages environmental geometric characteristics to improve the accuracy of path loss predictions [13]. Additionally, Qi et al. conducted studies examining the impact of UAV vibrations on air-to-air channel models, incorporating key components such as line-of-sight (LOS), specular reflection, and diffuse scattering to enhance the model's accuracy and reliability [14]. Islam et al. proposed optimization techniques for UAV deployment aimed at maximizing communication coverage and throughput. Their work addresses critical challenges such as trajectory planning and resource allocation, ensuring efficient and effective utilization of UAVs in communication networks [15]. These advancements contribute to the effective integration of UAVs in wireless communication networks. Phunchongharn et al. presented a resource allocation approach for D2D communications in advanced LTE cellular networks, focusing on optimizing resource usage to enhance network efficiency and address challenges such as interference management and QoS assurance [16]. Tang et al. highlighted that interference management and QoS assurance remain significant challenges in cellular networks with D2D underlay. They identified operational mode selection—whether cellular, direct, or reuse mode—as a critical issue in optimizing network performance [17]. Doppler et al. emphasized that D2D communication often relies on the reuse of licensed spectrum resources, enabling efficient utilization of available bandwidth [18]. Deploying UAVs on spectrum bands shared with D2D networks introduces interference management challenges. In scenarios with limited or no infrastructure, recent studies have explored D2D communication using licensed spectrum [19], demonstrating improvements in coverage and capacity for wireless networks like cellular systems. Particularly in dense or public safety scenarios, D2D enables direct communication between users without substantial infrastructure. Literature also addresses the coexistence of D2D and cellular communication under a base station's coverage [20]. Mozaffari, et al. [21], the study examines energy efficiency in cellular network architecture with the aid of UAVs (drones) in the context of Device-to-Device (D2D) communication. One advantage of D2D communication in a cellular network using UAVs is reduced resource usage and decreased response time. However, interference management poses a significant challenge for deploying UAVs within the D2D underlay of cellular networks. Also, in [22] presents an analytical framework for coverage and data rate analysis, considering UAV mobility in the air to serve multiple ground devices. The analysis incorporates critical parameters such as UAV altitude, path loss, user density, communication link distance, and transmission power. Sabuj et al. propose a model for RF energy harvesting in cognitive radio networks using a two-slope path-loss model, analyzing outage probability, energy harvesting, and optimization, though real-world applicability may be limited [23]. In [24] investigates UAV-assisted relay networks in the THz band, optimizing channel capacity through forward and backward trajectories, demonstrating significant performance improvements but with limitations in specific environments. Compared to previous studies on the coexistence of D2D and cellular networks, such as [25, 26], the inclusion of an aerial UAV base station with D2D communication introduces new challenges: Channel Modeling: The channel between the UAV and ground users is no longer a classical fading channel but instead depends on probabilistic Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) links [27]. Meanwhile, the channel between a ground base station and users continues to follow a Rayleigh fading model. Adjustable Altitude: Unlike fixed ground base stations, the UAV's adjustable altitude influences channel characteristics and coverage performance. UAV Mobility: The UAV's potential mobility adds complexity to the analysis, requiring an evaluation of its impact on D2D and overall network performance. Previous studies on UAVs, such as [9, 28], have not addressed the third challenge related to mobility in-depth.

This study explores the role of drone-based base stations in enhancing wireless network performance with device-to-device (D2D) communication, particularly in scenarios where fixed ground base stations are unavailable. Unlike prior research that models networks with random topology using Poisson Point Process (PPP), this study addresses the dynamic

topology introduced by drone mobility, highlighting unique challenges such as interference management, energy-efficient resource allocation, and variable network characteristics influenced by drone altitude and limited energy. By focusing on drone-specific constraints and real-world dynamics, the research proposes tailored algorithms for power and interference control, advancing beyond static PPP-based analyses. It evaluates the performance of drone-equipped networks in public safety scenarios, aiming to establish reliable communication between cellular and D2D users during emergencies, while addressing critical issues like optimal bandwidth allocation, energy efficiency, and interference mitigation in highly dynamic environments.

2 Mathematical models

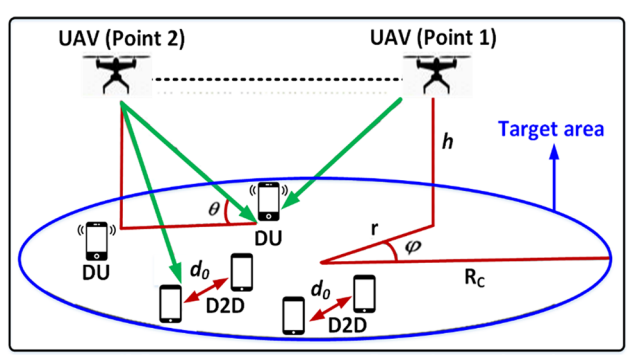
2.1 Network model

In this scenario, a circular region with radius *RC* is defined, which accommodates a number of wireless users. A drone, acting as a Low Altitude Platform (LAP), is deployed within this region to serve a subset of these users. The focus is on downlink communications from the drone, where Device-to-Device (D2D) users act as a secondary communication layer. The network users are categorized into two groups: cellular users (denoted as d_u) and D2D users. Typically, the average number of users within a specified area is computed as the product of user density and the total area size. The system model is illustrated in Fig. 1 [9].

The complexity of the proposed model arises from several factors, including the Poisson Point Process (PPP) distribution of cellular and D2D users, which requires integration over the spatial area to account for signal reception and interference. The interference from the drone and other D2D transmitters introduces additional complexity, as each receiver experiences interference from an effectively infinite number of transmitters, leading to a non-trivial interference calculation. The computational complexity grows with the number of users and transmitters, requiring O(N) calculations for each receiver, where N is the number of users or transmitters, and O(N^2) for network-wide interference analysis. As user density and network size increase, the system's capacity and interference modeling become more challenging, necessitating approximation techniques like Monte Carlo simulations to handle large-scale networks efficiently. The cellular users within the cell are uniformly distributed according to a Poisson Point Process (PPP) with a density λ_{du} (number of users per square meter). In Fig. 2, cellular users are distributed within a cell with a radius of 700 m and a density of 5×10^{-5} users per square meter.

The D2D transmitter users within the cell are assumed to be uniformly distributed following a Poisson Point Process (PPP), with a density of λ_d (users per square meter). As depicted in Fig. 3, D2D transmitters are scattered within a cell of radius 600 m, with a density of 5×10^{-5} users per square meter. In this model, it is assumed that each D2D transmitter has a fixed distance to its corresponding receiver, which is positioned in an isotropic direction [29]. This means that for each D2D communication pair, the transmitter and receiver are separated by a constant distance, which could be, for example, 10, 20, or 30 m. Importantly, the receivers themselves are not distributed according to a Poisson Point Process. Instead, they are assumed to be located at a fixed distance from their respective transmitters. Each receiver, therefore, experiences interference from an effectively infinite number of D2D transmitters located throughout the cell. This assumption is standard in PPP-based analyses, where the interference from transmitters is modeled as a random process over a large spatial area. It ensures that the average interference level perceived by a receiver remains constant, regardless of the specific location of either the transmitter or the receiver within the cell

Fig. 1 System model



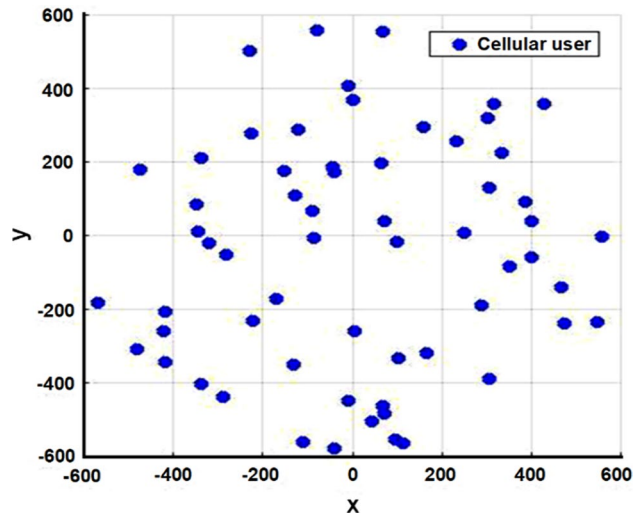


Fig. 2 The ppp distribution of cellular users

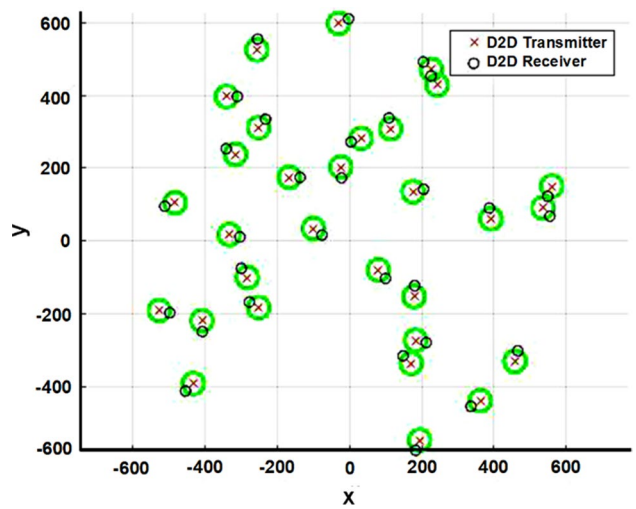


Fig. 3 D2D users ppp distribution

[30–33]. This approach simplifies the analysis by treating interference as a spatially homogeneous process, which is common in many network modeling scenarios.

2.2 SINR D2D

The received signal at the D2D receiver consists of three main components: the desired signal from the corresponding D2D transmitter pair, interference from the drone, and interference from other D2D transmitters within the cell. Specifically, the desired signal is the one intended for the D2D receiver, originating from its associated transmitter. The interference from the drone arises due to the simultaneous communication occurring between the drone and its assigned cellular or D2D users within the same area. Additionally, interference from other D2D transmitters is caused by the overlapping coverage of multiple transmitters within the network, leading to unwanted signals that may degrade the quality of the desired communication. Equation (1) of the received signal formulation highlights the primary sources of interference that impact the performance of the D2D communication system. Understanding the contributions of these interference components is essential for accurately evaluating the signal quality and system capacity.

$$\gamma_d = \frac{p_{r,d}}{I_d^c + I_{u+N}} \tag{1}$$

The received power $P_{r,d}$ is the desired signal strength from the D2D transmitter, while I_d^c represents interference from other D2D transmitters. I_u is interference from the drone, and N is the noise power at the receiver. Together, these components determine the overall signal quality, as described in Eq. (1).

$$p_{r,d} = p_d d_0^{-a_d} g_0 \tag{2}$$

where, d_0 represents the fixed distance between the D2D transmitter and receiver, while P_d denotes the transmission power of the D2D transmitter. a_d is the path loss factor for D2D connections, which accounts for signal degradation over distance. g_0 and g_i are the channel gains, where g_0 is the gain between the corresponding transmitter–receiver pair, and g_i represents the gains between all transmitters and receivers in the network. It is assumed that the power of all D2D transmitters and the distance between the desired D2D pairs are identical for all D2D connections, as expressed in Eq. (2).

$$\begin{aligned} I_d^c &= \sum\nolimits_{i \neq 0} P_d d_i^{-a_d} g_i \\ I_d &= \sum\nolimits_i P_d d_i^{-a_d} g_i \end{aligned} \tag{3}$$

The index $i = 0$ is used to refer to the selected D2D pair. Based on Eq. (2) and Eq. (3), it can be concluded that Eq. (4) represents the relationship between the parameters, incorporating the specific D2D pair and the associated power, distance, and channel gains.

$$I_c^d + P_{r,d} = I_d \tag{4}$$

where, I_u represents the interference received from the drone, which introduces an air-to-ground interference link. As the drone operates at a low altitude, it is essential to account for the probability of LoS conditions between the drone and the receiver. The probability of LoS depends on factors such as the relative positions of the drone and receiver, atmospheric conditions, and any obstacles that might block the signal. Incorporating this probability helps to more accurately model the interference and signal quality in the system.

$$\begin{aligned} P_{Los_{prim}} &= \frac{1}{1 + C\exp(-B[\theta - C])} \\ I_u &= P_u g_u \\ g_u &= P_{Los_{prim}} \left| X_{u_{prim}} \right|^{-a_u} + \eta P_{NLos_{prim}} \left| X_{u_{prim}} \right|^{-a_u} \end{aligned} \tag{5}$$

In this expression, $\left| X_{u_{prim}} \right|$ represents the distance between the D2D receiver and the drone. P_u is the transmission power of the drone, which determines the strength of the signal it emits. *P*LOS is the probability of a LoS between the drone and the D2D users, which is influenced by factors such as relative positioning and environmental conditions. η is an additional attenuation factor used to model the signal loss in NLoS conditions, where obstacles may obstruct the direct path between the drone and the receiver. For D2D communications, the channel is assumed to undergo Rayleigh fading, which is a common model for wireless channels in environments with multipath propagation [34–36]. This fading reflects the rapid variation in signal strength due to the presence of multiple reflecting paths.

2.3 Cellular users' SINR

The Cellular Users' Signal-to-Interference-plus-Noise Ratio (SINR) SINR*du* for users who can connect to the UAV is calculated using Eq. (6). This SINR represents the ratio of the received signal power from the UAV to the total interference and noise power, allowing for the assessment of the communication quality between the cellular user and the UAV. The calculation takes into account both the desired signal strength and the interference from other users and noise, providing a measure of the user's ability to maintain a reliable connection with the UAV.

$$\gamma_{du} = \frac{P_{r,u}}{I_d + N} \tag{6}$$

Here, I_d represents the interference from all D2D transmitters, and $P_{r,u}$ is the received signal power from the UAV. The ground receiver can receive signals from three different types: Line-of-Sight LoS, NLoS, and multipath-reflected signals. The likelihood of each type of signal is influenced by factors such as building density, height, and the elevation angle of the drone. LoS and NLoS signals are typically modeled separately, with multipath fading often neglected due to its lower probability. NLoS links tend to experience higher path loss because of shadowing and reflections from surrounding obstacles. This results in additional path loss factors beyond the free-space path loss, affecting both LoS and NLoS connections. The received signal power at the user's location is therefore determined by whether the connection is LoS or NLoS [37, 38].

$$P_{r,u} = \begin{cases} P_u |X_u|^{-a_u} LosLink \\ \eta P_u |X_u|^{-a_u} Nloslink \end{cases} \tag{7}$$

In this scenario, P_u is the transmission power of the UAV, and $|X_u|$ is the distance between the cellular user and the UAV. a_u represents the path loss factor from the cellular user to the UAV, and η is an additional attenuation factor applied in NLoS connections. The probability of a LoS connection, denoted as P_{LOS}, is influenced by various factors including the environment, building density, building height, the positions of the user and UAV, and the elevation angle between them. To calculate the received power at the cellular user from the UAV, the LoS probability P_{LOS} must be determined. The LoS probability can be expressed using the following equation (Eq. 8) [37].

$$P_{Los} = \frac{1}{1 + C\exp(-B[\theta - C])} \tag{8}$$

where, C and B are constant values that vary depending on the environmental context, such as rural, urban, or dense urban settings. The parameter θ represents the angle between the ground user and the UAV's altitude, measured in degrees. As illustrated in Fig. 1, if we consider three points—the location of the UAV, the location of the cellular user, and the projection of the UAV onto the ground—these points form the vertices of a triangle. The relationship between these points can be expressed by Eq. (9), which incorporates the variables C, B, and θ.

$$\sin\theta = \frac{h}{|X_u|} \Rightarrow \theta = \sin^{-1}\frac{h}{|X_u|} \tag{9}$$

As indicated by the formula, an increase in the elevation angle θ between the user and the UAV leads to a higher probability of LoS communication. This is because a larger elevation angle typically means the UAV is positioned more directly above the user, reducing the likelihood of obstructions (such as buildings or other obstacles) that could block the direct path between the two, thereby increasing the chance of LoS.

2.4 Coverage probability of D2D user

Consider a D2D receiver located at (r, φ), where r is the radius and ϕ is the angle (the angle between the UAV's ground projection and the center of the cell) in a polar coordinate system. Assuming the UAV is positioned at the center of the target area, which ensures the highest coverage probability, the coverage probability for a D2D receiver at location (r, φ), which connects to its D2D transmitter at a distance d_0, can be expressed using Eq. (10). This coverage probability depends on the receiver's location within the area, the distance to the corresponding D2D transmitter, and the UAV's position at the center.

$$\begin{aligned} P_{\mathrm{cov},d}(r,\varphi,\beta) &= \exp\left(\frac{-2\pi^2\lambda_d\beta^{2/}a^d d_0^2}{a_d\sin(\frac{2\pi}{a_d})} - \frac{\beta d_0^{a_d} N}{P_d}\right) \\ &\times\left[P_{Los}\exp\left(\frac{-\beta d_0^{a_d} P_u |X_u|^{-a_u}}{P_d}\right) + P_{NLos}\exp\left(\frac{-\beta d_0^{a_d} P_u |X_u|^{-a_u}}{P_d}\right)\right] \end{aligned} \tag{10}$$

Given that the UAV creates interference for D2D users, increasing the UAV's altitude to increase its distance from D2D users does not necessarily reduce interference for D2D users. As demonstrated in the simulations below, with an increase in UAV altitude, the D2D coverage probability initially decreases and then increases. This phenomenon arises because, as indicated by Eq. (8) and Eq. (9), although increasing the UAV altitude causes greater path loss, it also leads to a higher probability of LoS communication. Typically, D2D users prefer a NLoS view towards the UAV and a maximum distance from it; however, these two objectives conflict with each other. Furthermore, increasing the D2D transmission power P_d consistently enhances the D2D coverage probability, even in scenarios where interference is minimal (i.e., $N = 0$, with noise neglected). The D2D coverage probability, as expressed in Eq. (10), decreases with an increase in UAV transmission power. To address this issue, D2D users can either increase their transmission power or reduce the fixed distance parameter $d0$. Moreover, decreasing the density of D2D users improves coverage probability by reducing interference. It is important to note that the result presented in Eq. (9) pertains to the coverage probability for a D2D user at location (r,ϕ). To calculate the average coverage probability across the entire cell, a uniform distribution of users over the area is considered, with the user density described by $f(r,\varphi) = r/\pi R_c^2$. The average coverage probability of D2D users is then expressed in Eq. (11).

$$\begin{aligned}\bar{P}_{\mathrm{cov},d}(\beta) &= \mathrm{E}_{r,\varphi}[P_{\mathrm{cov},d}(r,\varphi,\beta)] = \exp\left(\frac{-2\pi^2\lambda_d\beta^{2/a_d}d_0^2}{a_d\sin\left(\frac{2\pi}{a_d}\right)} - \frac{\beta d_0^a N}{P_d}\right)\\ &\times\int_{\circ}^{R_C} E_{lu}\left[\exp\left(\frac{-\beta d_{\circ}^{alfa_d} I_u}{P_d}\right)\right] f(r,\varphi)drd\varphi = \exp\left(\frac{-2\pi^2\lambda_d\beta^{2/a_d}d_0^2}{a_d\sin\left(\frac{2\pi}{a_d}\right)} - \frac{\beta d_0^a N}{P_d}\right)\\ &\times\int_{\circ}^{R_C} E_{lu}\left[\exp\left(\frac{-\beta d_{\circ}^{alfa_d} I_u}{P_d}\right)\right]\frac{2r}{R_C^2}dr\end{aligned} \tag{11}$$

The average coverage probability for D2D users increases as the area size R_C grows. This occurs because, when the UAV serves a larger area, the average distance between the D2D users and the UAV increases. As a result, D2D users typically experience less interference from the UAV. The coverage probability of D2D users is defined as the probability that the values of γ_d (the signal-to-interference-plus-noise ratio, SINR) exceed the SINR threshold β. This relationship captures the effectiveness of the system in ensuring that the signal strength at the D2D receivers is strong enough to maintain reliable communication, even in the presence of interference.

2.5 Cellular users coverage probability

The coverage probability of a cellular user located at coordinates (*r*, φ) is given by the probability that the SINR (Signal-to-Interference-plus-Noise Ratio) exceeds a certain threshold β. This probability depends on the received signal power from the UAV, the interference from other cellular users and D2D transmitters, and the noise level. The formula for the coverage probability can be expressed as Eq. (12).

$$\begin{aligned}P_{\mathrm{cov},d}(r,\varphi,\beta) &= P[\gamma_{du} \geq \beta] = P_{Los}(r)\mathrm{P}\left[\frac{P_u r^{-a_u}}{I_{d+N}} \geq \beta\right] + P_{NLos}(r)\mathrm{P}\left[\frac{\eta P_u r^{-a_u}}{I_{d+N}} \geq \beta\right]\\ &= P_{Los}(r)\mathrm{P}\left[I_d \leq \frac{P_u r^{-a_u} - \beta N}{\beta}\right] + P_{NLos}(r)P\left[I_d \leq \frac{\eta P_u r^{-a_u} - \beta N}{\beta}\right]\end{aligned} \tag{12}$$

where, γ_{cell} is the SINR for the cellular user, β is the SINR threshold required for successful communication, *r* and ϕ represent the polar coordinates of the cellular user within the area of interest. It is noteworthy that there is no closed-form expression for the cumulative distribution function (CDF) of D2D user interference [38–41]. Therefore, here, we present the lower and upper bounds for the CDF of interference and divide the interference of D2D transmitters into two subsets. According to Eq. (12), the coverage probability is divided into two sets: upper bound (Eq. (13), Eq. (14)) and lower bound (Eq. (15), Eq. (16)).

$$\overline{P}^{L}_{\text{cov,du}}(\beta \cdot h) = \int_0^{R_c} P_{\text{LOS}}(r \cdot h) L_1\left(\frac{P_u |X_u|^{-\alpha_u}}{\beta} - N\right)\frac{2r}{R_c^2}dr + \int_0^{R_c} P_{\text{NLOS}}(r \cdot h) L_1\left(\frac{\eta P_u |X_u|^{-\alpha_u}}{\beta} - N\right)\frac{2r}{R_c^2}dr \tag{13}$$

$$L_1(T) = \left[1 - \frac{2\pi\lambda_d \Gamma\left(1 + \frac{2}{\alpha_d}\right)\left(\frac{T}{P_d}\right)^{-\frac{2}{\alpha_d}}}{\alpha_d - 2}\right] \times \exp\left(-\pi\lambda_d\left(\frac{T}{P_d}\right)^{-\frac{2}{\alpha_d}}\Gamma\left(1 + \frac{2}{\alpha_d}\right)\right) \tag{14}$$

Were, $\Gamma(t) = \int_0^{\infty} x^{t-1}e^{-x}dx(t)$.

$$\overline{P}^{U}_{\text{cov,du}}(\beta \cdot h) = \int_0^{R_c} P_{\text{LOS}}(r \cdot h) U_1\left(\frac{P_u |X_u|^{-\alpha_u}}{\beta} - N\right)\frac{2r}{R_c^2}dr + \int_0^{R_c} P_{\text{NLOS}}(r \cdot h) U_1\left(\frac{\eta P_u |X_u|^{-\alpha_u}}{\beta} - N\right)\frac{2r}{R_c^2}dr \tag{15}$$

$$U_1(T) = \exp\left(-\pi\lambda_d\left(\frac{T}{P_d}\right)^{-\frac{2}{\alpha_d}}\Gamma\left(1 + \frac{2}{\alpha_d}\right)\right) \tag{16}$$

In Eq. (14) and Eq. (16), when the density of D2D users approaches infinity$\lambda_d \Rightarrow \infty \Rightarrow U_1(T) = L_1(T) = 0 \Rightarrow P_{\text{cov},du} = 0$ Therefore, cellular users receive unlimited interference from D2D users, which causes the coverage probability for cellular users to become zero. Additionally, as the density of D2D users increases, the average coverage probability for du users decreases. In this case, to improve the coverage performance for du users, either the transmission power of the UAV (P_u) should be increased or R_c should be reduced. In the case where the density and power of D2D users are low, the interference received from D2D users is negligible compared to the interference received from the UAV, so it can be ignored. In this case, the average coverage probability for cellular users, which is at its maximum value, is equal to Eq. (17).

$$\overline{P}_{\text{cov,du}}(\beta) = \int_0^{R_c} \min\left[\left(\frac{P_u}{\beta N}\right)^{\frac{1}{\alpha_u}}, R_c\right] P_{\text{LOS}}(r)\frac{2r}{R_c^2}dr + \int_0^{R_c} \min\left[\left(\frac{\eta P_u}{\beta N}\right)^{\frac{1}{\alpha_u}}, R_c\right] P_{\text{NLOS}}(r)\frac{2r}{R_c^2}dr \tag{17}$$

The coverage probability for cellular users is defined as the probability that the signal-to-interference-plus-noise ratio (SINR), denoted as γ_{du}, exceeds the SINR threshold β. This threshold represents the minimum required SINR for successful communication, and the coverage probability reflects the likelihood that the cellular user experiences sufficient signal quality to maintain a connection.

2.6 Total system throughput

Now, the achievable rate for users d_u and D2D is examined, which can be expressed as Eq. (18) [30].

$$C_{du} = W\log_2(1 + \beta)\overline{P}_{cov,du}(\beta)$$
$$C_d = W\log_2(1 + \beta)\overline{P}_{cov,d}(\beta) \tag{18}$$

The average total system throughput $\overline{C}_{sum}$ can be computed by considering the coverage probabilities for both cellular users (d_u) and D2D users, as well as the number of users in each category. This total throughput is a function of the coverage probability and the density of the users. Assuming $u = \lambda_{du/\lambda d}$, the system throughput can be expressed as Eq. (19).

Table 1 Simulation parameters

Parameter Title	Symbol	Value
Transmission Power of Drone	P_u	5 w
Transmission Power of D2D Transmitter	P_d	100 mw
Path Loss Factor	K	30 dB
Path Loss for Cellular Users	a_u	2
Path Loss for D2D Users	a_d	4
Noise Power	N	−120 dBm
Bandwidth	W	1 MHz
Fixed Distance Between D2D Pair	d_0	20 m
Loss Factor for NLos	η	20 dB
Urban Environment Density Parameter	B,C	11.95. 0.136

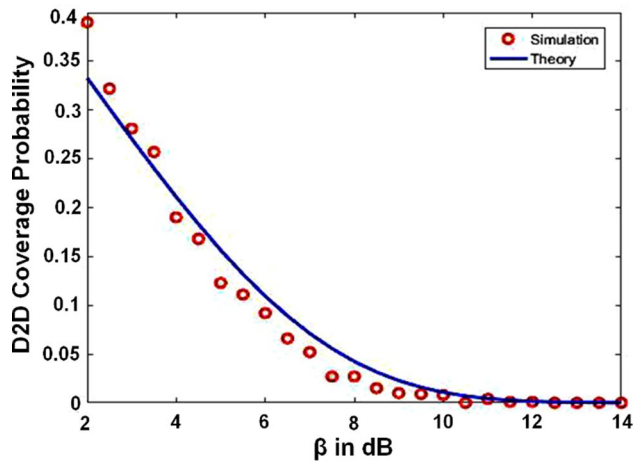


Fig. 4 D2D coverage probability

$$\overline{C}_{sum} = \lambda_d R_c^2 \pi \left[\mu \overline{P}_{cov,du}(\beta) + \overline{P}_{cov,d}(\beta) \right] W \log_2(1+\beta) \tag{19}$$

From Eq. (19), it can be concluded that the total system throughput $\overline{C}_{sum}$ is directly proportional to the density of cellular users (λ_u), but it also depends on the coverage probabilities of both the cellular ($P_{cov,du}$) and D2D users ($P_{cov,2du}$). Both of these coverage probabilities are decreasing functions of the D2D user density (λ_d). Therefore, while increasing λ_u (the density of cellular users) will directly increase the throughput, increasing λ_d (the density of D2D users) does not necessarily lead to an increase in the overall system rate. This is because the interference caused by a higher density of D2D users negatively affects the coverage probability, which in turn limits the achievable throughput for both types of users. Hence, there is a trade-off between user density and system performance.

3 Result and discussion

The simulation parameters are specified in Table 1. The coverage probability of D2D users based on the threshold β is obtained in Fig. 4.

According to Eq. (13) and Eq. (15), increasing R_C decreases the average coverage probability for *du* users. However, with the increase in R_C, according to Eq. (11), the average coverage probability for D2D users increases. Figure 5 shows the coverage probability of cellular users.

Figure 6 illustrates the total system throughput (in Gbps) as a function of the SINR threshold β for a bandwidth of 1 MHz, a drone altitude of 500 m, and two different values of λ_d (D2D user density). As shown in the figure, for high values of β, the received SINR cannot exceed the threshold, leading to coverage probabilities approaching zero. According to Eq. (18), while $log_2(1+\beta)$ increases with an increase in β, the coverage probability decreases exponentially, whereas $log_2(1+\beta)$ increases only logarithmically. This results in the average system rate approaching zero for large values of

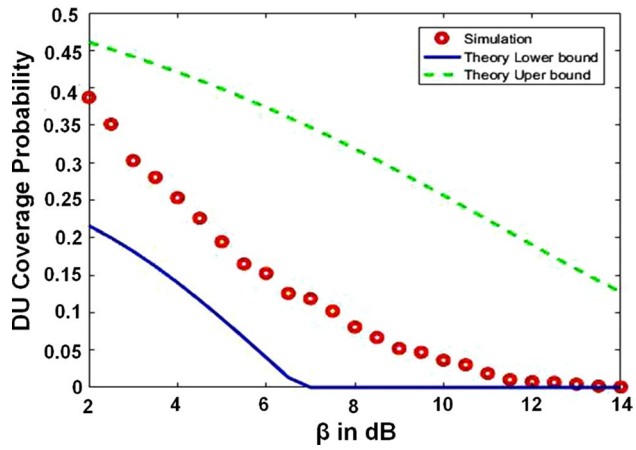


Fig. 5 The probability of cellular user coverage

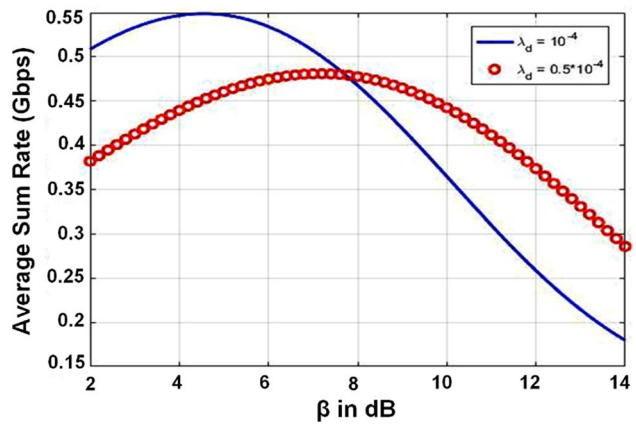


Fig. 6 The total system rate based on the SINR threshold

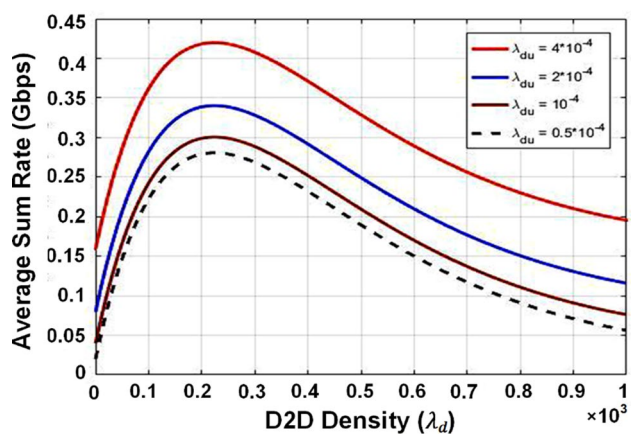


Fig. 7 The total system rate based on λ_d and for different values of λ_{du}

β. Conversely, when the threshold β approaches zero, $log_2(1+\beta)$ also tends to zero, causing the system throughput to approach zero, while the coverage probabilities approach 1. Thus, there is a trade-off between the SINR threshold and system throughput, where the throughput decreases for both high and low values of β.

Figure 7 illustrates the impact of D2D density (λ_d) on the total system rate for different values of λ_{du} (density of cellular users). Generally, reducing the number of D2D users in a region also decreases the system throughput rate. A high D2D density reduces the coverage probability, thus decreasing the data rate for each user. However, since the total system rate is directly proportional to the number of D2D users, increasing D2D density can also improve the total system rate. According to Fig. 7, as the density of users at the cell edge increases, the optimal λ_d that maximizes the total rate decreases. This is because, with an increase in λ_{du}, the contribution of cellular users to the total system rate increases. Thus, increasing the rate for each cellular user leads to an overall increase in system throughput. To boost the rate for cellular

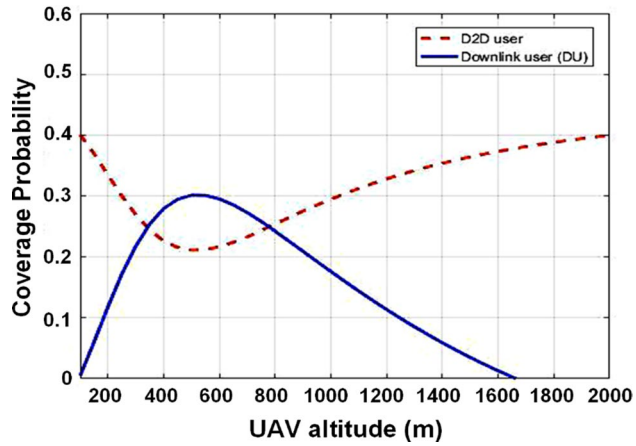


Fig. 8 Optimal UAV altitude

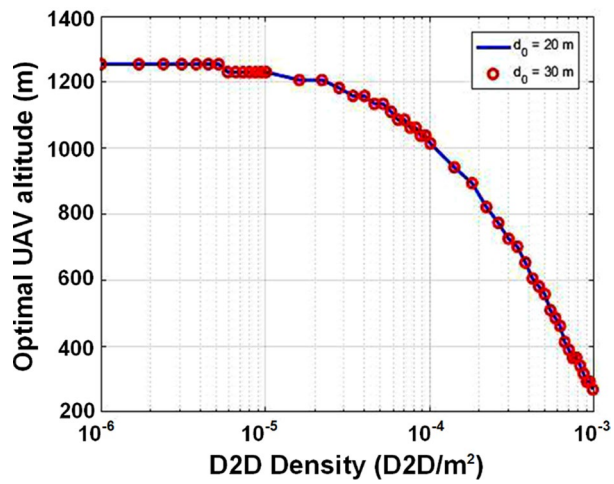


Fig. 9 Optimal UAV altitude based on λ_d

users, the number of D2D users, which act as interference sources for cellular users, should be reduced. Consequently, the optimal λ_d decreases as λ_{du} increases.

Considering Eq. (8), Eq. (15), and Eq. (17), it is evident that increasing the UAV altitude (h) can enhance both the probability of LoS and the coverage probability. However, as the altitude increases, the distance $|X_u|$ also increases, which causes both L_1 and U_1 (from Eq. (16) and Eq. (18)) to decrease. This reduction in L_1 and U_1 results in a decreased coverage probability for users at lower altitudes. Therefore, to maximize the coverage probability, the UAV's altitude must be carefully adjusted to strike a balance between improving LoS probability and minimizing the path loss caused by increased distance.

Figure 8 illustrates the trade-off between the coverage probability of cellular (d_u) and D2D users as a function of UAV altitude. Cellular users experience the best coverage at moderate UAV altitudes, around 500 m, where the balance between LoS probability and path loss is optimal. However, at very low altitudes, cellular users suffer from significant shadowing and interference, while at high altitudes, although LoS improves, the path loss increases due to the large distance from the UAV. For D2D users, the interference from the UAV is more pronounced at low altitudes, but as the altitude increases, the UAV's interference diminishes, and D2D users achieve maximum efficiency at very high altitudes. Figure 9 shows that the optimal altitude for maximizing cellular user coverage increases with D2D user density, as higher D2D density leads to more interference, necessitating a higher UAV altitude to mitigate this. This highlights the need for dynamic altitude adjustment to balance the conflicting needs of cellular and D2D users for optimal coverage.

As seen in Fig. 9, the optimal UAV altitude for downlink users decreases as the density of D2D users increases. This occurs because a higher D2D user density result in increased interference for downlink users, prompting the UAV to lower its altitude to improve the Signal-to-Interference-plus-Noise Ratio (SINR) for these users. In essence, the UAV moves closer to downlink users to mitigate the interference caused by the higher density of D2D users. Interestingly, the optimal altitude of the UAV remains unaffected by the fixed distance d_0 between the D2D transmitter and receiver. This is because the distance between

D2D users does not influence the interference level experienced by downlink users, meaning changes in d_0 do not impact the UAV's optimal altitude. Figure 10 further illustrates how the total system rate varies with UAV altitude for different values of d_0, showing that the system's performance is not sensitive to variations in the fixed distance between D2D users.

The optimal UAV altitudes that maximize the total system rate are approximately 200, 400, and 500 m for distances d_0 of 20, 25, and 30 m, respectively. The optimal altitude, which leads to the highest system rate, is influenced by the density of both cellular and D2D users. For example, when $d_0 = 20$, the total system rate begins to increase when the UAV altitude exceeds 1300 m. This increase occurs because the coverage probability for cellular users approaches zero, and the system throughput becomes primarily determined by D2D users. As the UAV altitude rises, the interference between D2D users decreases, which results in an improvement in the D2D user coverage probability ($\overline{C}_d$), contributing to the increase in the total system rate.

4 D2D communications (uplink, downlink, and public safety scenarios)

To complete a telecommunications system, a comprehensive performance analysis is crucial, considering both the uplink and downlink communication channels. This analysis should account for factors such as coverage probability, SINR, system throughput, and user density. Furthermore, during public emergencies, ensuring the prompt and reliable delivery of critical messages is paramount. A robust system must be able to prioritize emergency traffic, maintain minimal latency, and guarantee reliable communication, even in scenarios with high user density and limited resources. Such a system should also be resilient to interference, whether from other users, environmental factors, or network congestion, ensuring continuous and efficient operation during times of crisis.

4.1 SINR in uplink scenario

The power of the D2D transmitters and the distance between D2D transmitter–receiver pairs are assumed to be the same for all pairs in the system. This assumption simplifies the analysis by treating the D2D communication links as homogeneous, allowing for a more straightforward evaluation of the system's performance. By assuming uniform transmission power and fixed distances, the impact of interference from D2D users on the cellular network and the overall system throughput can be analyzed more easily, as variations in these parameters are not considered. However, in practical scenarios, these factors may vary, and further refinement of the model could be necessary to capture the real-world complexity of D2D communication networks.

$$I_u = \sum_j P_{du} r_{d_{2d}-d_{uj}}^{-\alpha_{cell}} g_{uj} \Rightarrow if \rightarrow a_d = a_{cell} \rightarrow I_u = \sum_j P_{du} r_{d_{2d}-d_{uj}}^{-\alpha_d} g_{uj} \tag{20}$$

where, P_d is Power of the D2D transmitter, u a variable that could represent the number of users or an additional parameter in the summation, r_{d2d} the distance between the D2D transmitter and receiver, d_{uj} the distance between the D2D receiver and a cellular user, a_{cell} path loss exponent for cellular communication, g_{uj} channel gain or a related factor for the cellular communication link. The simulation parameters are specified in Table 2.

The Success Transmission Probability (STP) for uplink connections is a key metric for evaluating the performance of communication systems, particularly during critical situations such as natural disasters. STP measures the likelihood

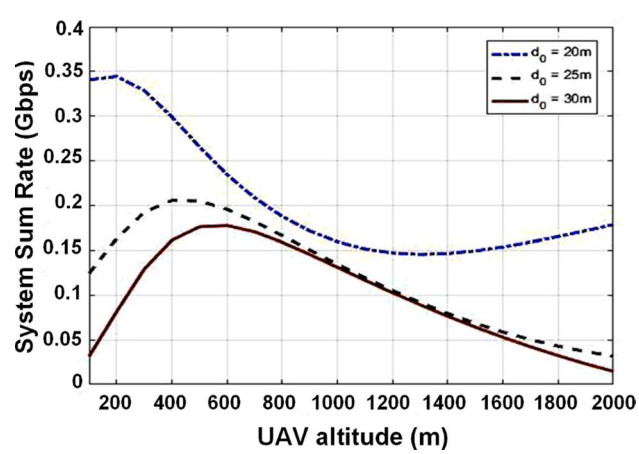


Fig. 10 The total system rate based on the drone altitude

that a transmission from a device to a receiver will be successful, which is crucial for ensuring the reliable delivery of messages. Just like the coverage probability in downlink scenarios, the STP is a probability value, where the sum of the probabilities of successful and unsuccessful transmissions equals one. The STP for uplink D2D connections is generally determined by factors such as SINR, transmission power, and the distance between the D2D transmitter and receiver. The formula for calculating STP, as given by Eq. (21), would take these factors into account to provide an assessment of the likelihood of successful transmission under given conditions.

$$\begin{aligned} STP_d(\beta) &= P\left[\gamma_d \geq \beta\right] = \exp\left(-\zeta(\lambda_d + \lambda_{du}\left(\frac{P_{du}}{P_u}\right)^{\frac{2}{a_d}} - \frac{\beta d_0^{a_d} N}{P_d}\right) \\ \zeta &= \pi\beta^{\frac{2}{a_d}} d_0^2 \Gamma(1+\frac{2}{a_d})\Gamma(1-\frac{2}{a_d}) \end{aligned} \tag{21}$$

The equation for the Success Transmission Probability (STP) in Eq. (21) highlights the interaction between various network parameters and the probability of successful transmission at a D2D receiver. As the density of both cellular users (λ_{du}) and D2D users (λ_d) decreases, the likelihood of successful transmission increases. This is primarily due to a reduction in interference at the D2D receiver, which is beneficial for ensuring clear communication. Interference from cellular users, however, increases as their transmission power rises, leading to a lower STP. To counter this, increasing the transmission power of the D2D sender can improve the probability of successful transmission by enhancing the signal strength at the receiver. Additionally, Eq. (22) considers the situation where a drone receives communication from a distant cellular user at coordinates (r, ϕ). This relationship helps in quantifying how various factors, including transmission power and distance, influence the probability of successful communication in a drone-based network.

$$STP_{du}(\beta) = P\left[\gamma_{du} \geq \beta\right] = P_{Los}\mathrm{I}\left(\frac{P_{du}X_u^{au} - \beta N}{\beta}\right) + P_{NLos}\mathrm{I}\left(\frac{\eta P_{du}X_u^{-a_u} - \beta N}{\beta}\right) \tag{22}$$

According to Eq. (22), the parameters of the drone and its communication channels with cellular users play a crucial role in determining the probability of successful transmission. Specifically, the drone's altitude and the characteristics of the communication channel affect both the LoS probability P_{Los} and the NLoS probability PN_{Los}. As the drone's altitude increases, the LoS probability P_{Los} increases, resulting in a higher likelihood of direct communication. However, this also leads to a decrease in the function (·) for both the upper and lower bounds, which subsequently reduces the overall probability of successful transmission. In Fig. 11, the impact of these parameters is illustrated, showing how the probability of successful transmission for cellular users changes with variations in the drone's altitude. The figure clearly indicates that as the altitude increases, while the LoS improves, the overall transmission success rate may decline due to the geometric and interference factors involved.

Table 2 Simulation parameters

Parameter title	Symbol	Value
Transmission power of drone	P_u	5 w
Transmission power of cellular users	P_{du}	250 mw
Transmission power of D2D transmitter	P_d	100 mw
Path loss factor	K	30 dB
Path loss for cellular users	a_u	2
Path loss for D2D users	a_d	4
Path loss between cellular users and D2D users	a_{cell}	3.5
Noise power	N	−120 dBm
Bandwidth	W	1 MHz
Fixed distance between D2D pair	d_0	20 m
Loss factor for NLos	η	20 dB
Urban environment density parameter	B,C	11.95. 0.136

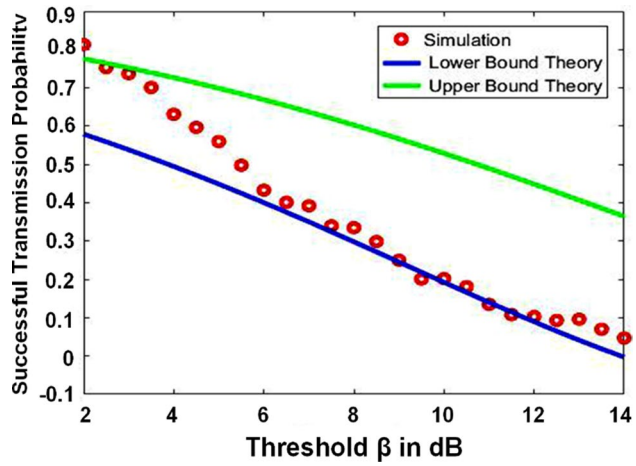


Fig. 11 Successful Transmission Probability for cellular users

4.2 Downlink scenario

The total energy efficiency is defined as the ratio of the total system sum rate to the average number of users, which is determined by the transmission power of the D2D sender and the drone. Specifically, the average number of users is given by $\pi R_c^2 \lambda_d$ for cellular users and $R_c^2 \lambda_d$ for D2D users. Energy efficiency depends on the average user rate, which in turn is influenced by the probability of successful transmission. As the number of D2D users increases, the total energy efficiency improves, but beyond a certain optimal point, it starts to decline. This behavior is explained by the logarithmic relationship in $log_2(1+\beta)$, where increasing the threshold β increases $log_2(1+\beta)$, but also reduces the probability of successful transmission, leading to diminishing returns. Figure 12 demonstrates the total energy efficiency for different values of λ_d, with the results specified for $\lambda_{du}=10^{-4}$. The figure shows that initially, increasing λ_d enhances energy efficiency, but after reaching an optimal density, further increases in λ_d lead to a reduction in energy efficiency due to the trade-off between increased interference and reduced transmission success.

As the density of cellular users increases, the interference they generate for D2D users also increases. This heightened interference results in a decrease in the probability of successful transmission for D2D users, thereby lowering the overall system performance. Consequently, the total energy efficiency, as shown in Fig. 13, decreases with the rise in cellular user density. The increased interference leads to a reduction in the average user rate for D2D users, and despite the increase in the number of users, the overall energy efficiency suffers due to the negative impact on transmission success. Therefore, managing cellular user density is crucial for optimizing the total energy efficiency in D2D-based systems.

Figure 14 illustrates the total energy efficiency for various values of cellular user density, as a function of D2D user density. As shown in the figure, when the D2D user density exceeds $\lambda_d=5\times10^{-4}$, the total energy efficiency approaches zero. This decline occurs due to the significant interference caused by the high density of D2D users. As the number of D2D users increases, the interference they generate for both cellular and other D2D users escalates, leading to a

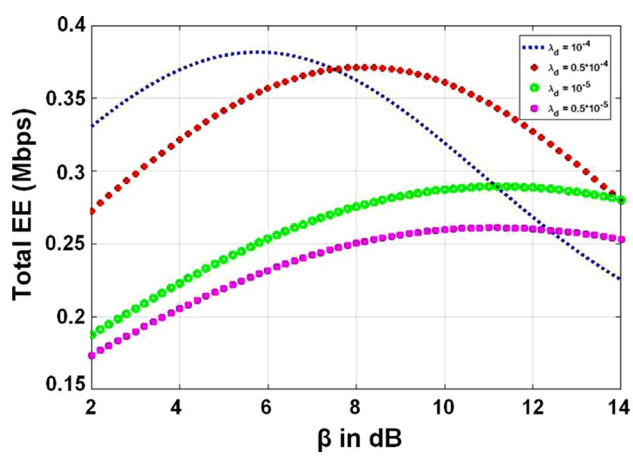


Fig. 12 Total energy efficiency for different values of λ_d

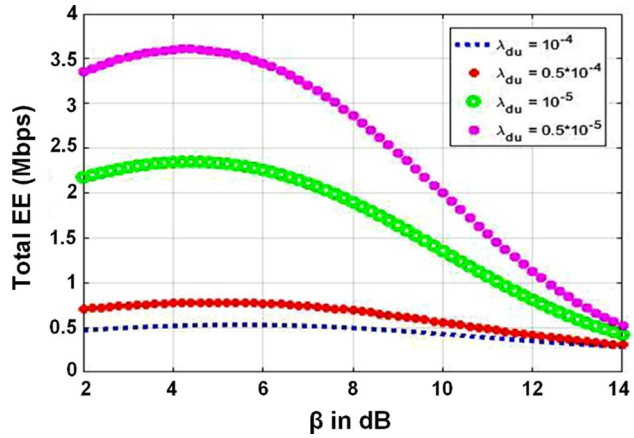


Fig. 13 Total energy efficiency for different values of λ_{du}

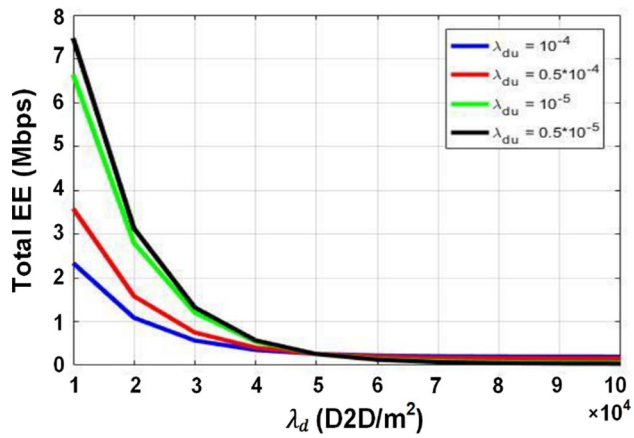


Fig. 14 The total energy efficiency based on λ_d and for different values of λ_{du}

reduction in the overall system performance. Consequently, the energy efficiency is severely impacted when D2D user density becomes too high, highlighting the need for careful management of D2D user density to maintain optimal system performance.

4.3 Public safety scenario

In emergency situations, where rapid communication is essential, the current system setup allows for D2D users, who typically do not have a direct communication link with the drone or cellular users, to send critical messages. In the underlay mode, D2D users usually do not have their own dedicated frequency band but instead share the available spectrum with cellular users. This situation is particularly challenging when D2D users need to communicate with the drone or other users in real-time. To overcome this limitation, a direct link is established between D2D users and cellular users, allowing D2D users to indirectly connect to the drone through the cellular users in downlink mode. This approach enables timely message delivery and helps ensure connectivity in critical situations, where efficient communication can be vital for disaster management or other urgent needs.

The SINR of cellular users is related to the drone connection, while other cellular users are transmitting data for D2D receiver users. Therefore, a cellular user experiences interference from other cellular users who are transmitting data for D2D users (Eq. (23)).

$$\gamma_{du} = \frac{P_{ru}}{I_d + I_u + N} \tag{23}$$

P_{ru} is the received power at cellular users, I_d is the interference received from all D2D transmitting users, I_u is the interference from other cellular users who are transmitting data to D2D receivers, and N is the noise power.

$$\begin{aligned} P_{Los} &= \frac{1}{1 + C\exp(-B[\theta - C])} \\ Los_{Link} &= P_{Los} P_u X_u^{-a_u} \\ NLos_{Link} &= P_{NLos} \eta P_u X_u^{-a_u} \\ P_{ru} &= Los_{Link} + NLos_{Link} \\ I_d &= \sum\nolimits_i P_d d_i^{-a_d} g_i \end{aligned} \tag{24}$$

where d_i is the distance from the D2D transmitter to the cellular user, and d_j is the distance between cellular users. The coverage probability of cellular users is calculated in Eq. (25).

$$\begin{aligned} P_{\mathrm{cov},d_u} &= P_r[\gamma_u \geq \beta] = P_r[\frac{P_{ru}}{I_d + I_u + N} \geq \beta] = P_{Los}(r)\mathrm{P}[\frac{P_{ru} r^{-a_u}}{I_d + I_u + N} \geq \beta] + P_{NLos}(r)\mathrm{P}[\frac{P_{ru} r^{-a_u}}{I_d + I_u + N} \geq \beta] \\ &= P_{Los}(r)\mathrm{P}[I_d + I_u \leq \frac{P_{ru} r^{-a_u} - \beta N}{\beta}] + P_{NLos}(r)\mathrm{P}[I_d + I_u \leq \frac{P_{ru} r^{-a_u} - \beta N}{\beta}] \end{aligned} \tag{25}$$

It is worth mentioning that there is no closed-form expression for the cumulative distribution function (CDF) of the interference of D2D users. Therefore, upper and lower bounds for the CDF of the interference are provided.

$$\begin{cases} \phi_1 = \left\{ \phi_B \middle| I_d + I_u \geq T \right\} \\ \phi_2 = \left\{ \phi_B \middle| I_d + I_u \leq T \right\} \end{cases} \tag{26}$$

where T is the threshold used to obtain the CDF of D2D user interference. In this case, we have defined a new path loss exponent called a_{cell}, which differs from αd and a_u, but its value is close to a_d. Assuming the independence of interferences I_d and I_u in Eq. (27).

$$\begin{aligned} &\mathbb{P}[I_d + I_u \leq T] \cong \\ &\mathbb{E}_{d_i,d_j}\left[\prod_{i\in\Phi_b} \mathbb{P}\{P_d d_i^{-\alpha_d} g_i \leq T\} \prod_{j\in\Phi_c} \mathbb{P}\{P_{du} d_j^{-\alpha_{cell}d} g_j \leq T\}\right] \\ &= \mathbb{E}_{d_i}\left[\prod_{i\in\Phi_b} \mathbb{P}\left(g_i \leq \frac{T d_i^{-\alpha_d}}{P_d}\right)\right] \mathbb{E}_{d_j}\left[\prod_{j\in\Phi_c} \mathbb{P}\left(g_j \leq \frac{T d_j^{-\alpha_{cell}}}{P_{du}}\right)\right] \\ &\approx^{(a)} \mathbb{P}\left[\prod_{i\in\Phi_b}\left(1 - \exp\left(-\frac{T d_i^{\alpha_d}}{P_d}\right)\right)\right] \mathbb{P}\left[\prod_{j\in\Phi_c}\left(1 - \exp\left(-\frac{T d_j^{\alpha_{cell}d}}{P_{du}}\right)\right)\right] \\ &\approx^{(b)} \exp\left(-\lambda_d \int_0^\infty \left(1 - \exp\left(-\frac{T d_i^{\alpha_d}}{P_d}\right)\right) r dr\right) \exp\left(-\lambda_{cell} \int_0^\infty \left(1 - \exp\left(-\frac{T d_j^{\alpha_{cell}}}{P_{du}}\right)\right) r dr\right) \\ &= \exp\left(-\pi\lambda_d \left(\frac{T}{P_d}\right)^{\frac{2}{\alpha_d}} \Gamma\left(1 + \frac{2}{\alpha_d}\right)\right) \exp\left(-\pi\lambda_u \left(\frac{T}{P_{du}}\right)^{\frac{2}{\alpha_{cell}d}} \Gamma\left(1 + \frac{2}{\alpha_{cell}d}\right)\right) \end{aligned} \tag{27}$$

Parts (a) and (b) follow the characteristics of Rayleigh fading channels and the probability function of a Poisson Point Process (PPP), respectively.

$$\begin{aligned}\mathbb{P}[I_d + I_u \geq T|\phi_1 = 0] &\leq^{(C)} \frac{E\left[I_d \geq T|\phi = 0\right]}{T} + \frac{E\left[I_u \geq T|\phi = 0\right]}{T} \\ &= \frac{1}{T}E\left[\sum_{\phi} P_d d_i^{-a_d} g_i \parallel (P_d d_i^{-a_d} g_i \leq T)\right] + \frac{1}{T}E\left[\sum_{\phi} P_{du} d_i^{-a_{celld}} g_i \parallel (P_{du} d_i^{-a_{celld}} g_i \leq T)\right]\end{aligned} \tag{28}$$

where c is based on Markov's inequality, which is expressed as follows: For any non-negative random variable X and positive L, it is equal to Eq. (29).

$$\mathbb{R}(X \geq L) \leq \frac{E[X]}{L} \tag{29}$$

Additionally, ||(.) is the indicator function, which can only take the values of 1 or 0. Lower Bound and Upper Bound functions are presented as in Eq. (30). Probability of cellular user coverage in a public safety scenario shown in Fig. 15.

$$\begin{aligned}L(T) &= \left(1 - \frac{2\pi\lambda_d\Gamma(1+2\alpha_d)}{\alpha_d - 2}(\frac{T}{Pd})^{-\frac{2}{\alpha_d}} - \frac{2\pi\lambda_u\Gamma(1+2\alpha_{cell_d})}{\alpha_{cell_d} - 2}(\frac{T}{Pdu})^{-\frac{2}{\alpha_{cell_d}}}\right) \\ &\times \exp\left(-\pi\lambda_d(T_P pd)^{-\frac{2}{\alpha_d}}\Gamma(1+\frac{2}{\alpha_d})\right) \times \exp\left(-\pi\lambda_u(T_{pdu})^{-\frac{2}{\alpha_{cell_d}}}\Gamma(1+\frac{2}{\alpha_{cell_d}})\right) \\ U(T) &= \exp\left(-\pi\lambda_d^{-\frac{2}{\alpha_d}}\Gamma(1+\frac{2}{\alpha_d})\right) \times \exp\left(-\pi\lambda_u(T_{pdu})^{-\frac{2}{\alpha_{cell_d}}}\Gamma(1+\frac{2}{\alpha_{cell_d}})\right)\end{aligned} \tag{30}$$

In a system where D2D users and cellular users alternate as the desired signal source for other D2D users, two modes must be considered when calculating the SINR: the Desired D2D Mode and the Cellular Users Desire Mode. In the Desired D2D Mode, the D2D pair transmits the desired signal to each other over a Rayleigh fading channel, with the SINR depending on the transmission power, interference, and noise levels. In the Cellular Users Desire Mode, cellular users transmit to the D2D users, with the SINR affected by the cellular user's transmission power, the distance between the cellular user and D2D receiver, and interference from both D2D and cellular users. The coverage probability of D2D users in the Desired D2D Mode, expressed by Eq. (31), quantifies the likelihood of successful signal reception above a specified SINR threshold, taking into account user density, interference, and system parameters like the SINR threshold and Rayleigh fading characteristics.

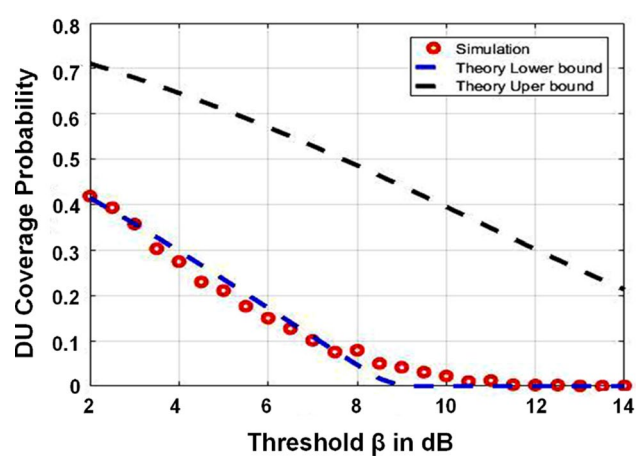


Fig. 15 Probability of cellular user coverage in a public safety scenario

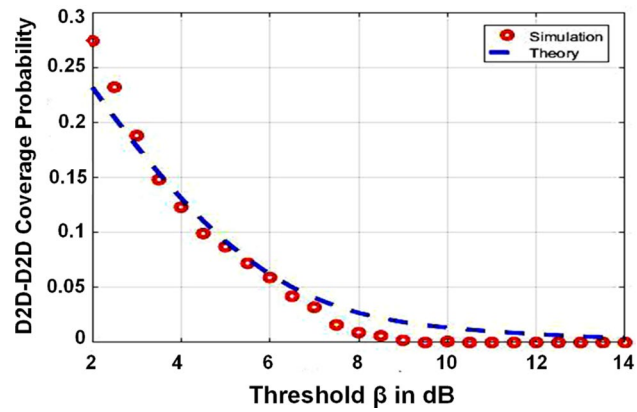


Fig. 16 Coverage probability of D2D users when the desired signal is from a D2D pair

$$
\begin{aligned}
&\mathbb{P}_{\text{cov - }dd2d}[r,\phi,\beta] = P[\gamma_{dd2d} \geq \beta] = \mathbb{P}\left[P_d d_0^{-\alpha_d} g \geq \left(I_u + I_{\text{cell - }d2d} + I_d + N\right)\beta\right] \\
&= \mathbb{P}\left[g \geq \frac{\beta d_0^{\alpha_d}(I_u + I_{\text{cell - }d2d} + I_d + N)}{P_d}\right] \\
&= (a)\mathbb{E}_{I_u, I_{\text{cell - }d2d}, I_d}\left[\exp\left(-\frac{\beta d_0^{\alpha_d}(I_u + I_{\text{cell - }d2d} + I_d + N)}{P_d}\right)\right] \\
&= (b)\mathbb{E}_{I_u}\left[\exp\left(-\frac{\beta d_0^{\alpha_d} I_u}{P_d}\right)\right]\mathbb{E}_{I_{\text{cell - }d2d}}\left[\exp\left(-\frac{\beta d_0^{\alpha_d} I_{\text{cell - }d2d}}{P_d}\right)\right] \\
&\times\mathbb{E}_{I_d}\left[\exp\left(-\frac{\beta d_0^{\alpha_d} I_d}{P_d}\right)\right]\exp\left(-\frac{\beta d_0^{\alpha_d} N}{P_d}\right)
\end{aligned}
\tag{31}
$$

In this scenario, the variable g follows an exponential distribution with a mean of one (g~exp(1)), which models the fading conditions of the channel, particularly Rayleigh fading. The interference, including for I_u, $I_{cell-d2d}$ and I_u, is treated as random variables, contributing to the overall system performance. As shown in Fig. 16, the coverage probability for D2D users when the desired signal is from a D2D pair decreases, primarily due to the interference from cellular users. The interference from cellular users affects the SINR of D2D users, thereby reducing their coverage probability, which emphasizes the negative impact of increased interference in the system.

In the "Cellular Users Desire" mode, the desired signal originates from a cellular user. This communication mode aims to facilitate a connection between cellular users and D2D users, allowing the D2D users to establish indirect communication with the drone via cellular users. This setup enables D2D users to benefit from cellular infrastructure, ensuring that they can still maintain communication with the drone even if direct communication between D2D users

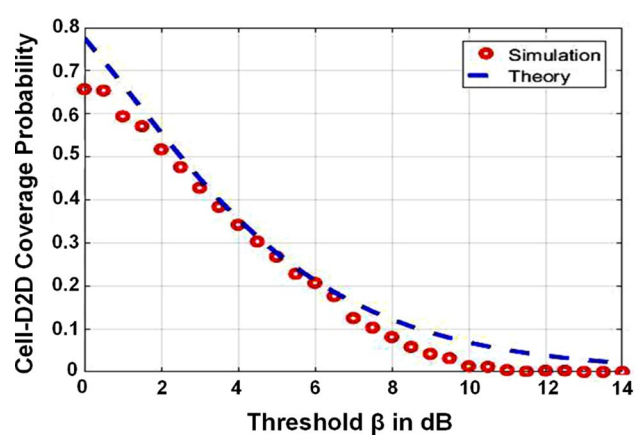


Fig. 17 Coverage probability of D2D users when the desired signal is from a cellular user

and the drone is not possible. The system is designed to allow D2D users to leverage cellular network resources in emergency situations, enhancing connectivity and coverage.

$$
\begin{aligned}
&\gamma_{\text{dcell}} = \frac{P_{\text{cell}} r}{I_u + I_d + N} \\
&\mathbb{P}_{\text{cov - dcell}}[r, \phi, \beta] = P[\gamma_{\text{dcell}} \geq \beta] = \mathbb{P}\left[\frac{P_{\text{cell}} r}{I_u + I_d + N} \geq \beta\right] \\
&T = \beta(I_u + I_d + N) \\
&\mathbb{P}[P_{\text{cell}} r \geq T] = 1 - \mathbb{P}[P_{\text{cell}} r \leq T] \\
&= 1 - \mathbb{E}_{d_i, g_i, T}\left[\prod \mathbb{P}\left(P_{\text{cell}} d_i^{-\alpha_{\text{cell}}} g_i \leq T\right)^{\Phi_c}\right] \\
&= 1 - \mathbb{E}_{d_i, g_i, T}\left[\prod \mathbb{P}\left(g_i \leq \frac{T d_i^{\alpha_{\text{cell}}}}{P_{\text{cell}}}\right)^{\Phi_c}\right] \\
&= 1 - \mathbb{E}_{d_i, T}\left[\prod \left(1 - \exp\left(-\frac{T d_i^{\alpha_{\text{cell}}}}{P_{\text{cell}}}\right)\right)_{i \in \Phi_c}\right]
\end{aligned}
\tag{32}
$$

As shown in Fig. 17, there is a high coverage probability, primarily due to the fact that, in public safety conditions, all cellular users are attempting to send a single message to the receiving D2D users. The large number of cellular users increases the likelihood of successful transmission, as they collectively help ensure the delivery of the message to D2D users. The higher the number of cellular users involved in the communication process, the greater the probability of successfully reaching the D2D receivers, thereby improving overall system reliability and coverage in emergency scenarios.

5 Conclusion

This paper thoroughly examines the performance of Device-to-Device (D2D) communication in drone-assisted cellular networks, focusing on key metrics such as coverage probability, system throughput, and energy efficiency. The results reveal that increasing UAV altitude improves line-of-sight communication, thereby enhancing coverage probability. However, the benefits of higher altitudes are counterbalanced by increased path loss, leading to a reduction in overall system throughput. The analysis also identifies an optimal D2D user density of 50 users per km^2, which effectively minimizes interference and improves resource utilization. Furthermore, interference plays a significant role in the success transmission probability, which decreases by 25% when the SINR threshold increases from 0 to 10 dB. In the context of public safety scenarios, the study highlights the advantages of underlay in-band communication, which enables D2D users to establish reliable indirect links via cellular users, ensuring the successful delivery of critical messages while also improving energy efficiency. These findings emphasize the potential of drone-based D2D communication networks in enhancing coverage, flexibility, and reliability, particularly for emergency response and natural disaster recovery efforts. Moving forward, further research could explore dynamic UAV altitude adjustment strategies, adaptive resource allocation algorithms, and the integration of machine learning techniques to optimize network performance in real-time. Additionally, investigating the impact of environmental factors such as weather conditions and urban topography on drone-assisted D2D communication would provide valuable insights for improving network resilience in challenging scenarios.

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Declarations

Competing interests The authors declare no competing interests.

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